Electrical & Electronic Engineering (EEE) and Computer Engineering (CE)
Final Year Projects (2026-2027)

Project Title: Title: An Intermittent Computing In-Ear Basal Body Temperature Sensor

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	Control	X	Embedded Systems		High Frequency Electronics		Microelectronics
	Electric Power	X	Software	X	Connected Health		MEMS
	Cyber-Security		Wireless Communications		Signal/Image Processing	X	Intelligent Systems
	Digital Design		Sensor Networks		Data Analytics	X	Electronics

Continuous and reliable temperature monitoring can provide valuable information for both healthcare and occupational safety. This project will develop an **in-ear basal body temperature (BBT) sensing system** capable of monitoring and transmitting body temperature using a compact, low-power embedded platform.
One potential healthcare application is the monitoring of patients with physical wounds. Continuous temperature measurements could provide an indication of changes in physiological condition and potentially support the **early detection of abnormal temperature trends associated with infection or sepsis**. The system could also be investigated for occupational applications, particularly for workers exposed to high-temperature environments, such as **firefighters and industrial workers**, where continuous temperature monitoring may help identify conditions associated with **heat stress**.
The proposed system will investigate the use of an in-ear temperature sensor as a non-invasive approach to monitoring body temperature. While similar devices are already available commercially, the key research aspect of this project is to investigate whether such a system can operate using **energy harvested from its surrounding environment**, for example through light, kinetic energy, or radio-frequency (RF) energy. This approach aims to reduce or eliminate reliance on conventional batteries that require replacement or recharging once depleted.
The system will initially be developed and evaluated using an **MSP430FR5994-powered testbed**, together with a compatible temperature sensor. The project will investigate how sensing, computation, and communication can be designed to minimise energy consumption while maintaining sufficient measurement accuracy and system functionality. Particular attention will be given to operation under **intermittent power availability**, where the harvested energy may not be continuously available.



Figure 1: Fruit picking robot

This project will require completion of a number of tasks including but not exclusively the following:

Expected Tasks

1. **Conduct a literature review** and investigate the principles and requirements of basal body temperature measurement using an in-ear sensing approach. The basic functionality of the system will be to monitor temperature over a specified timespan and, if the measured temperature exceeds a predefined normal range, relay a warning signal to the wearer or monitoring system.
2. **Develop an experimental testbed** for evaluating the selected temperature sensor, including measurement accuracy, response time, power consumption, and operation under different conditions.
3. **Develop software to support intermittent computing.** The software should provide the functionality required of the sensor described in the first task and implement suitable baseline techniques for surviving or

mitigating power failures. The implementation will be guided by existing approaches reported in the literature rather than developing such techniques from scratch.
4. **Evaluate the developed system** in terms of temperature measurement accuracy, energy efficiency, communication performance, and reliability.

Learning Outcomes
Upon completion of the project you will expect to have:
1. Developed an understanding of **wearable and connected-health sensing systems**.
2. Gained practical experience with **temperature sensors and embedded-system development**.
3. Developed knowledge of **low-power and intermittent computing techniques**.
4. Developed hands-on experimental and laboratory skills involving the characterisation of performance and energy-efficiency of **hardware, firmware, and sensor data acquisition**.
5. Developed programming and data-analysis skills for **processing and evaluating sensor measurements**.

MEng Project
If assigned as an MEng EEE/CE project, the scope of work will be extended and include additional tasks and learning outcomes. MEng extensions must be discussed and agreed with the Supervisor and Moderator. Relevant suggestions for extensions are:
1. **Investigating options for integrating wireless communication**, particularly **Bluetooth and LoRaWAN**, considering factors such as **communication range, data rate, latency, reliability, and energy consumption**. The aim will be to determine the suitability of different communication approaches for an energy-constrained, intermittently powered wearable sensor.
2. Investigate the application of **transprecision computing** to reduce computational and energy requirements while maintaining an acceptable level of temperature measurement accuracy. This extension will explore whether computational precision can be adapted according to the requirements of the application rather than always using unnecessarily high numerical precision. Such an approach may reduce computational and energy costs while preserving the accuracy required for reliable temperature monitoring.